House Price Prediction

This project focuses on predicting house prices using regression-based machine learning models. The dataset was preprocessed using Pandas, where categorical features were encoded and data was prepared for modeling. The data was then split into training and testing sets in an 80:20 ratio.

Two models were implemented: Linear Regression and Random Forest Regressor. Their performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. The Linear Regression model achieved an R^2 score of approximately 0.65, with an average prediction error of around 9–10 lakhs, indicating moderate accuracy in capturing overall pricing trends.

The analysis revealed that area is the most influential feature affecting house prices, followed by the number of bedrooms, bathrooms, and key amenities such as air conditioning and parking. These features contribute significantly to the variation in property prices.

Interestingly, Linear Regression outperformed Random Forest in this case, suggesting that the relationship between features and house prices is largely linear and does not require complex non-linear modeling. This highlights that simpler models can sometimes be more effective when the underlying data relationships are straightforward.

Based on these findings, it is recommended that real estate businesses prioritize property size and essential amenities when estimating house prices. The model can serve as a useful baseline tool for making consistent and data-driven pricing decisions.

Overall, this project demonstrates a complete machine learning workflow, from data preprocessing to model evaluation, and provides practical insights into factors influencing house prices.